The Living Line — Cheat Sheets (all nineteen, one printable file)

Companion to the guide. One sheet per lesson. Print this file, or screenshot the individual sheets in cheatsheets/ for your phone. Each sheet re-defines its own jargon — no memory required. Links: see the guide's Lesson 19 (every one verified 16 Aug 2026; the pixel-art links of Sheets 15–18 on 20 Aug 2026).

Sheet 0 · The Universal Order (works for every drill)

> **Cause first → body next → loose parts last → settle.**
>
> Pocket version: **C·B·L·S — "Cool Bodies Lag Softly."**

The four questions, before frame one:

- ☐ What CAUSES this motion? (muscle / wind / magic / impact)
- ☐ What LEADS? (eyes / hips / chest / hand)
- ☐ What LAGS? (hair / sleeves / cape / belly)
- ☐ What does the TIMING say? (timing = how many frames it takes — quick reads eager or nervous, slow reads heavy or grand)

Persona quick-dials:

- ☐ Quick timing = alert, young, nervous · Slow = heavy, calm, grand
- ☐ Even cushioned spacing (spacing = distance moved per frame) = elegant · Lurching = scruffy
- ☐ Big anticipation (wind-up before the action) = theatrical · Tiny = professional · NONE = alien
- ☐ Loose-part lag: crisp = controlled · floppy = warm mess · none = cardboard
- ☐ Symmetry = formal or uncanny · Asymmetry = alive

Sheet 1 · Timing & Spacing (Lesson 1)

Words on this sheet: **frame** = one single image; film shows 24 per second · **timing** = how many frames an action takes (the clock) · **spacing** = how far the drawing moves per frame

(the handwriting) · **slow in / slow out** = bunching drawings at the start and end so the move eases instead of snapping · **on twos** = each drawing held 2 frames.

> **Timing is the clock. Spacing is the handwriting.**

The bouncing-ball proof (do this first, always):

- ☐ Even spacing every frame = conveyor belt. Mechanical. Dead.
- ☐ Bunched at the arc's top, wide near the ground = hang… then SMACK. Gravity appears.
- ☐ Big bounces decaying into small ones = energy loss. Rubbery. Real.
- ☐ Tight quick bounces = nervous ping-pong ball · Long slow arcs = weary bowling ball. Same circle, different soul.

Easing dials:

- ☐ Deep easing both ends = calm, deliberate, graceful
- ☐ No easing at all = robotic, eerie, alien (useful on purpose!)
- ☐ Ease only the stop = startled, jumpy

Budget: 12 fps, one drawing per frame = the classic 24-fps-on-twos rhythm with easier arithmetic.

Sheet 2 · The Twelve Principles (Lesson 2)

Where they come from: Disney veterans **Frank Thomas & Ollie Johnston**, *The Illusion of Life* (1981). They are levers, not laws — every one is a personality dial.

The chant (twelve beats, in order — say it until it says itself):

> **Squish · wind-up · stage it · plan-or-don't · lag · ease · curve · fidget · clock it · push it · weigh it · charm.**

Beat	Principle	Personality lever
Squish	**Squash & stretch** — flatten under impact, lengthen with speed, volume constant	lots = bouncy, comic · none = stiff, dignified
Wind-up	**Anticipation** — small prep move before the action	big = theatrical · tiny = professional · none = alien
Stage it	**Staging** — present the action so it reads instantly	framing tells the audience how to feel
Plan-or-don't	**Straight-ahead / pose-to-pose** — discover frame by frame, or plan keys first	wild energy vs deliberate control
Lag	**Follow-through & overlap** — loose parts trail; nothing starts/stops in unison	the difference between alive and cardboard

Beat	Principle	Personality lever
Ease	**Slow in / slow out** — bunch drawings at both ends of a move	deep = graceful · none = mechanical, eerie
Curve	**Arcs** — natural motion travels curved paths	smooth arcs = elegance · jagged = chaos, panic
Fidget	**Secondary action** — small supporting action layered on the main one	a diary of who the character is
Clock it	**Timing** — how many frames an action takes	quick = eager, twitchy · slow = weary, grand
Push it	**Exaggeration** — push past real life so it reads	sets the register: naturalism → full cartoon
Weigh it	**Solid drawing** — forms with weight, volume, structure	credible from any angle
Charm	**Appeal** — order (clarity) + interest (surprise)	villains can drip with it too

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Sheet 3 · The Personality Toolkit (Lesson 3)

Words on this sheet: **silhouette** = the solid-black outline shape of a pose · **line of action** = the single imaginary curve running through a pose · **contrapposto** = weight on one leg, hips tilt one way, shoulders counter-tilt · **secondary action** = small supporting action layered on the main one.

Shape language (read before anything moves):

- ☐ Circles = soft, friendly, childlike
- ☐ Squares = sturdy, dependable, stubborn
- ☐ Triangles = sharp, dynamic, dangerous
- ☐ The dominant shape sets the first impression; motion confirms it — or betrays it *on purpose*

The two-second tests:

- ☐ SILHOUETTE: fill the pose with black — can you still tell what's happening? If mud-blob, redraw the pose, not the rendering
- ☐ LINE OF ACTION: draw the one curve first — proud back-arch = confident · collapsed C = defeated/sneaky · straight vertical = stiff, robotic

Asymmetry = alive:

- ☐ Weight on one leg (contrapposto), one arm wider, one eyebrow working harder
- ☐ Perfect symmetry is a special effect: soldiers, androids, porcelain aristocrats

The signature move:

- ☐ Give every character ONE repeatable motion habit (the "head swaggle" strategy) — spend it at emotionally right moments

The four questions: ☐ What causes it? ☐ What leads? ☐ What lags? ☐ What does the timing say?

Sheet 4 · Walk Cycle (Lesson 4)

Words on this sheet: **cycle** = drawings that loop seamlessly · **timing** = how many frames an action takes · **on twos** = each drawing held 2 frames.

Build order:

- ☐ CONTACT — both feet touch, legs widest, arms widest (this pose = 80 % of the walk's attitude)
- ☐ DOWN — lowest point, front knee bends, weight lands (heaviness lives here)
- ☐ PASSING — free leg passes planted leg, body rising
- ☐ UP — highest point, toes push off (bounce lives here)
- ☐ Repeat all four, other side → 8 drawings ≈ 16 frames on twos at 24 fps
- ☐ Hips lead · shoulders counter-rotate (twist opposite to hips)
- ☐ Arms opposite legs; hand lags forearm lags upper arm
- ☐ Head bobs gently, a beat behind the body
- ☐ Hair / hems / tail follow one beat later still
- ☐ Loop test: feet must not slide (onion skin = ghost view of neighbouring frames)

Recast without redrawing — turn dials only:

☐ Stride length ☐ Tempo ☐ Down-depth ☐ Up-height ☐ Head bob ☐ Arm swing ☐ Posture ☐ Symmetry

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Sheet 5 · Head Turn (Lesson 5)

Words on this sheet: **anticipation** = small prep move before the action · **overshoot & settle** = pass the target, ease back · **arc** = curved path of motion.

- ☐ EYES dart first (the thought arrives)
- ☐ ANTICIPATION — small dip or lean away
- ☐ TURN on an ARC — nose draws a rainbow, chin dips mid-turn
- ☐ Bunch drawings at both ends, wide spacing mid-turn (fast middle)
- ☐ Optional blink on the fastest frames
- ☐ OVERSHOOT a few degrees past the target

- ☐ SETTLE back
- ☐ Hair / earrings / collar arrive LATE and settle LAST
- ☐ Arc check: dot the nose every frame — dots must curve

Castings:

- ☐ dignified = long ease, quiet settle
- ☐ panicked = big anticipation + 3-frame snap
- ☐ haunted = eyes move LAST
- ☐ smug = add the 2-frame side-to-side "swaggle" at the end

Sheet 6 · Flight (Lesson 6)

Words on this sheet: **drag** = trailing part lags its base · **follow-through** = loose parts keep going after a stop · **squash/stretch** = compress under force, lengthen with speed.

First: what holds them up? ☐ wings (work) ☐ power (velocity) ☐ magic (invented rules)

Wings:

- ☐ A wing is an ARM (shoulder → elbow → wrist)
- ☐ Downstroke broad + powerful · upstroke folded + quick
- ☐ Feather tips always late (drag + follow-through)
- ☐ Body rides the flap, offset a beat; head steady; tail last

Power flight:

- ☐ Crouch (squash) → launch (stretch) — deeper crouch + fewer frames = more power
- ☐ Cruise on shallow arcs, cape streaming and rippling backwards
- ☐ Bank: eyes → body roll → cape sweeps wide
- ☐ Hover = tiny drift, never dead-still (dead-still = uncanny, on purpose only)
- ☐ Landing: bunch drawings, toe-first, knees swallow momentum — or the GROUND cracks instead (superhuman)

Swimming (water = thick sky):

- ☐ Undulator: the body IS the wave — one S-curve travels nose → tail (backwards along the body, forwards through the water)
- ☐ Tail-and-keel fish: firm body, sweeping tail, fins trail (drag); hovering = sculling, never dead-still
- ☐ Darter: coil → burst (2–3 frames) → long eased glide — the drag always wins
- ☐ Slow wide wave = serene/huge · fast tight wiggle = small/nervous · bubbles are the water's cape

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Sheet 7 · Human vs Alien vs Superhuman (Lesson 7)

Words on this sheet: **anticipation** = small prep move before the action · **slow in/out** = easing at the start and end of a move · **smear** = stretched/multiplied drawing held 1–2 frames to sell speed · **overlap** = parts start and stop at different times.

> **Every creature is a rule-set.** Human: familiar rules, imperfectly. Superhuman: familiar rules at impossible values, with control. Alien: *different* rules, kept with total consistency.

Human (all boxes ticked = a person):

☐ weight managed ☐ anticipation everywhere ☐ eases everywhere ☐ parts lead & lag ☐ nothing symmetrical ☐ arcs everywhere

Superhuman (keep human rules, push the values):

- ☐ 2–4 frame actions + smear, on clean arcs
- ☐ Tiny crouch, instant delivery, minimal recovery
- ☐ The WORLD reacts instead: ground cracks, dust storms, cape thunders — body stays composed
- ☐ Dead-clean stops; stillness with confidence

Alien (subtract 1–3 human rules — everywhere, always):

- ☐ No anticipation, ever = predatory
- ☐ Head before eyes = possessed
- ☐ Constant speed, dead stops = insectile
- ☐ Straight paths, angular turns = machine logic
- ☐ Everything arrives at once = puppet
- ☐ Loose parts inert (or moving FIRST) = horror
- ☐ No weight / wrong-way joints / perfect symmetry = elsewhere

Dosage: one break = uncanny · several = gloriously other. **Consistency:** the day your alien winds up before a jump, it becomes a person in a costume.

Sheet 8 · Elegance vs Scruffiness (Lesson 8)

Words on this sheet: **arc** = curved path of motion · **spacing** = distance moved per frame · **overshoot & settle** = pass the target, ease back · **boiling** = redrawn lines shimmering frame to frame — noise when accidental, texture when chosen.

> Two dialects of the same language. Neither is better. The only failure is the *accidental* version.

Dial	Elegant	Scruffy
Arcs	long, unbroken	kinked, jittery
Spacing	evenly graded, cushioned ends	lurches and hesitations
Timing	unhurried, consistent	syncopated; quick-quick-slow
Overshoot	barely-there, long soft settle	floppy, bouncy multi-hit settles
Follow-through	restrained, silk-like	everything flapping
Line of action	long, lifted	compressed, slumped
Silhouette	clean, simple	busy, spiky, delightfully messy
Line quality	few confident strokes	sketchy strokes, deliberate boiling
Symmetry	composed near-symmetry	cheerful dishevelment

Remember:

- ☐ Speed is NOT the dialect — a fencer's lunge is fast *and* elegant; a hungover shuffle is slow *and* scruffy
- ☐ The dialect lives in *quality of path and stop*
- ☐ Cross-cast for comedy: scruffy ballerina, elegant scarecrow

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Sheet 9 · Overlap & Follow-Through — hair, cloth, tails (Lesson 9)

Words on this sheet: **overlapping action** = parts start/stop at different times · **follow-through** = loose parts continue after the body stops · **drag** = trailing part lags its base.

- ☐ Animate the BASE first (head / shoulders / hips)
- ☐ Treat loose things as CHAINS: root → middle → tip
- ☐ Offset each segment 1–2 frames behind its parent
- ☐ Wave travels down the chain (S-curve mid-swing = you did it right)
- ☐ Tips travel farthest, overshoot most, settle last
- ☐ Body stops → loose parts keep talking for 6–10 more frames
- ☐ Decay of the swinging = the material's weight
- ☐ Hair = 2–4 chains max (fringe / sides / mass), roots welded to skull
- ☐ Materials: iron = short lag, fast decay · silk = long lag, slow decay
- ☐ Personality: crisp lag = controlled · floppy = warm mess · equal-everywhere = seaweed (fix it)

Sheet 10 · The Elements (Lesson 10)

Words on this sheet: **timing** = how many frames · **spacing** = distance per frame · **onion skin** = ghost view of neighbouring frames.

ELECTRICITY — no body:

- ☐ Jagged segments, sharp angles, branching forks
- ☐ Redraw a NEW random bolt every frame — ignore the onion skin (discontinuity = crackle)
- ☐ ~6 fps, 6 frames, glow layer ("Add (Glow)" mode in Clip Studio)
- ☐ Zero follow-through

WATER — all body:

- ☐ Simple readable masses, smooth muscular curves
- ☐ Crest lags base (overlap champion) · droplets = tiny balls on arcs
- ☐ Splash: impact → rim → columns/droplets (hang at top) → fall → late soft ripples

FIRE — buoyant appetite:

- ☐ Base leads, tips die: ASCENT → SWELLING → DIVISION → REDUCTION
- ☐ Never holds a shape; S-curves with sharp licks; few tongues, not scribble
- ☐ Candle = gentle sway · blaze = fast, big licks, sparks arrive late

WIND — invisible:

- ☐ Draw only its VICTIMS: hair, cloth, leaves, dust
- ☐ It's a traveling push: near things react BEFORE far things
- ☐ Gust → then recovery, each thing on its own schedule (overshoot, settle)
- ☐ Hair loop: 24 frames, keys every 3, softness over volume

Sheet 11 · Clip Studio Paint EX — quick path (Lesson 11)

- ☐ File > New → **Project: Animation** (set fps; 12 fps, or 24 on twos)
- ☐ Window > **Timeline** if hidden
- ☐ Drawings = **cels** inside an **animation folder**; New animation cel button to add
- ☐ **Onion skin** on: Animation menu > Show animation cels > Onion skin settings
- ☐ Inbetween with the **light table** (Animation cels palette — reference glows under your cel)
- ☐ Hold each drawing 2 frames = **on twos**; re-assign cels to reuse drawings (blinks, loops)

- ☐ Folders: BODY / LOOSE (1–2 frames late) / FX (glow = "Add (Glow)")
- ☐ Loop playback constantly — judge motion, never stills
- ☐ Export: File > **Export animation** → MP4 (share) or GIF (loops)
- ☐ EX = no frame cap (PRO/DEBUT stop at 24 frames)

Sheet 12 · ibisPaint X — quick path (Lesson 12)

- ☐ Open the **Animation bar** (frame strip at the bottom of the canvas)
- ☐ **+** adds a frame · long-press + drag reorders · duplicate/delete per frame
- ☐ **Duplicate-and-redraw** only what moved (volumes stay consistent)
- ☐ **Onion skin** on (adjust count + opacity of ghost frames)
- ☐ **Frame Settings**: fps + playback — **Loop** (cycles) / **Ping pong** (sway) / One-shot
- ☐ 6–8 frames, 8–12 fps is a full drill — fewer, stronger drawings
- ☐ Silhouette test: does the action read at thumbnail size?
- ☐ Export: Back button → **Save Animated Video**; watch it outside the app
- ☐ Squint test: if the action survives blur, the poses work

Sheet 13 · The Practice Program (Lesson 13)

Words on this sheet: **flour sack** = the classic practice prop — a half-full sack animated as a character; all acting, no face · **scrub** = step through a film frame by frame (VLC: the E key).

Session design (ADHD-honest):

- ☐ Decide the drill BEFORE opening the app (while the kettle boils)
- ☐ One sitting = one drill = 15–25 minutes. Done is the target
- ☐ Repeat drills shamelessly — the tenth pendulum teaches what the first couldn't
- ☐ Keep these sheets where you practise (the sheet remembers so your head can draw)
- ☐ End by watching your loop five times. Let it play. That's the reward

The weekly carousel (no score; climb on anywhere):

- ☐ Day 1 Ball → Day 2 Walk → Day 3 Head turn → Day 4 Overlap → Day 5 Element → Day 6 Flight → Day 7 Rewatch & remix ONE dial
- ☐ Always add a personality twist (tired / cocky / haunted / elegant / scruffy)
- ☐ Too easy? Do it as a flour sack · Gone stale? Do it at 16 px in 4–8 frames (Sheet 17)

The free film shelf (all public domain, links in the guide's Lesson 19):

- ☐ Fleischer *Superman* (1941–43) — capes, effects, flight
- ☐ *Gertie the Dinosaur* (1914) — personality through weight
- ☐ *The Sunshine Makers* (1935) — elegant vs scruffy in one film
- ☐ Preston Blair, *Advanced Animation* — copy one page per sitting

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Sheet 14 · Into the Game (Lesson 14)

Words on this sheet: **sprite** = a 2D image an engine can draw · **sprite sheet** = frames packed in one grid image · **flipbook** = a playable stack of frames · **fps** = frames per second — a property of your ART (8–12), not the game (60).

The pipeline (identical in every engine):

- ☐ Export transparent PNGs (hide the paper layer!)
- ☐ Zero-pad the names: `walk_01.png` … `walk_08.png` (engines sort alphabetically)
- ☐ Same canvas size + consistent pivot for every frame, or the character jitters
- ☐ Group frames per action (idle / walk / head_turn)
- ☐ Set the ANIMATION's fps — on twos = 12
- ☐ Loop cycles · one-shots fire a "finished" event → settle back to idle
- ☐ Crisp pixel art = Nearest/Point filtering · painterly = Linear/Bilinear

Where things live:

- ☐ GODOT: AnimatedSprite2D + SpriteFrames · `play("walk")` · `await sprite.animation_finished`
- ☐ UNITY: Sprite Editor (slice) → Animation window (clips) → Animator · `animator.Play("Walk")` · done when `normalizedTime >= 1`
- ☐ UNREAL: Sprite Actions → Extract Sprites → Create Flipbook · `SetFlipbook` + `PlayFromStart` · `OnFinishedPlaying`
- ☐ WEB: `<img src="loop.webp">` · CSS `steps(8)` · canvas `drawImage` player

Scene fades (Godot, one line):

- ☐ `await Fade.fade_out(0.8).finished` → change scene → `Fade.fade_in(0.8)`
- ☐ Universal Fade addon: copy folder, done — no autoload, 7 wipe patterns

The dial that survives: play-speed = the timing dial, live at runtime — ☐ Godot `speed_scale` ☐ Unity `animator.speed` ☐ Unreal `SetPlayRate` ☐ 0.6 = weary · 1.4 = eager

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Sheet 15 · The Grid — pixel fundamentals (Lesson 15)

Words on this sheet: **pixel** = the smallest square of colour; in pixel art, placed by hand · **cluster** = touching pixels of one colour (the real brushstroke) · **orphan pixel** = a lone pixel of its colour (noise unless deliberate) · **jaggies** = uneven stair-steps on a line · **doubles** = a 2-pixel corner on a 1-pixel line · **AA (anti-aliasing)** = in-between colours placed by hand at stair-steps · **selout** = an outline that lightens where lit.

> **P·I·X·L — "Palette small · Integer scale · eXport PNG · Look at 100 %."**

Before drawing:

- ☐ Automatic anti-aliasing OFF on pens, fills, selections, text (Lesson 18 has every menu)
- ☐ 1-pixel pen · 1-pixel grid · a second view at 100 %
- ☐ Start at 16×16, then 32×32 · pick the size as a casting choice (big head = cute; long body = serious)

While drawing:

- ☐ Think in clusters: big blobs first, edges last · hunt orphan pixels
- ☐ Lines: constant steps = straight (3-3-3) · consistently changing steps = curve (1-1-2-3-5) · anything else = jaggy
- ☐ Erase doubles (the L-corners) · lines are expensive — let contrast do their job
- ☐ AA only on steps longer than 1 px · never on 45° lines · never outward past a game sprite's edge · not at all ≤ 16 px
- ☐ Light source first → no pillow shading · break up banding
- ☐ Outline = a casting choice: black = toy/arcade · selout = lit/dimensional · none = painterly

Check at 100 %:

- ☐ Silhouette test (fill it black) · does one pixel of eye-shift change the mood? It should
- ☐ Angular clusters = hard/mechanical · rounded = soft/organic · stray orphans = scruffy (only if chosen)

Sheet 16 · Colour in Pixels (Lesson 16)

Words on this sheet: **palette** = the fixed list of allowed colours · **ramp** = dark→light colours for one material · **hue / saturation / value** = colour family / intensity / lightness · **hue-shifting** = hue moves cooler in shadow, warmer in light · **dithering** = two colours in a

pattern the eye blends · **indexed colour** = pixels store palette numbers, so editing the palette recolours the image · **palette swap** = one sprite, new colours, "new" character.

> **P·I·X·L — "Palette small · Integer scale · eXport PNG · Look at 100 %."** This sheet is the P.

Palette:

- ☐ Start with 4 colours (ARQ4) → 12 (AAP-Micro12) → 16 · 2–3 colours per body part · 6–12 per character
- ☐ Borrow from Lospec: ENDESGA 32 · DawnBringer 32 · PICO-8 · Sweetie 16 · AAP-64
- ☐ "If you can't make a good sprite in 4 colors, using 40 colors isn't going to help"

Ramps:

- ☐ Light source FIRST
- ☐ Shadow = value down + hue towards cool · Light = value up + hue towards warm · saturation: decide (PIXNOTE nudges it up in shadow, Medeiros down) and stay consistent
- ☐ Schlitter's ramps: ~20° of hue shift per swatch · value steps shrink near white · never 100 % saturation
- ☐ 16×16 → 2 tones per material · 64×64+ → 4–5
- ☐ Check in grey (desaturate, or a black layer on "Color" mode): if it still reads, the values work

Dither:

- ☐ Only between similar values · never on outlines · never the whole screen
- ☐ Best for big flat areas, banding, and texture that does double duty (stone, sand, weave)
- ☐ Heavy dither = gritty/retro/scruffy · flat clusters = clean/modern/elegant

The recolour trick (casting with colour only):

- ☐ Warm + contrast = hero · cool + desaturated = tired/ghost · dark + one hot accent = villain · pastels = gentle
- ☐ Indexed mode (Aseprite/Pixelorama): edit the palette, not the pixels

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Sheet 17 · Pixel Animation (Lesson 17)

Words on this sheet: **timing** = how long (frames or ms) · **spacing** = how far per frame — and in pixels the smallest step is a whole pixel · **ms per frame** = 100 ms = 10 fps · 125 ms = 8 fps · 83 ms ≈ 12 fps · 50 ms = 20 fps · **hold** = a frame shown longer · **sub-pixel animation** = moving an edge "half a pixel" by recolouring it, not moving it · **smear** = a

stretched drawing for 1–2 frames to sell speed · **Contact / Down / Passing / Up** = the four cornerstone drawings of a walk.

> **C·B·L·S — "Cool Bodies Lag Softly"** (cause → body → loose parts → settle) ·
> **P·I·X·L** — judge every loop at 100 % and real speed.

The spacing problem (1 px is a big step):

- ☐ Hold frames instead of sliding pixels · move a part, not the whole · sub-pixel recolour for slow motion · or embrace the bounce
- ☐ "Even 1 pixel of change is effective"

Frame budgets:

- ☐ 2 = flicker/blink · 3 = Mega Man run (A-B-C-B) · 4 = NES walk · 6 = walk with weight · 8 = fluid · 12+ = risk of floaty
- ☐ 8 frames @ 80 ms = 4 frames @ 160 ms (same clock, different handwriting)
- ☐ 8–12 fps standard · 4–6 fps idles · 15–24 fps attacks
- ☐ Hold the extremes ~2× longer than the inbetweens · one slipped pixel loops forever

Idle (the personality machine):

- ☐ 4 frames · body ±1 px · chest edge up · head a frame later · hair a frame after that
- ☐ Sub-pixel breath = calm · 2-px bob = nervous/eager · long hold at the bottom = heavy · eyes only = menace · swaggle on top = cocky

Walk / run / attack:

- ☐ 4 frames: Contact–Passing–Contact–Passing (body 1 px lower on Contact) · 6 adds Down · 8 has all four
- ☐ Head bob = triangle wave, not sine · legs first across all frames, then arms, head, torso, loose parts
- ☐ Dials: stride · bob (1 px poised, 2–3 px goofy) · arm swing (tight = stiff, full = peppy) · ms per frame · lean · asymmetry
- ☐ Run: airborne frame, lean, ~60 ms · Jump: crouch 1–2 → stretch → slow apex → squash (games often skip the crouch)
- ☐ Attack: load → smear at peak → HOLD the hit → 1–2 px follow-through → 1 px overshoot home (100/50/50/50/100/50 ms)

Loose parts & elements:

- ☐ Hair/cloth: 1 frame late · S-curve, crest moves 1 px to the tip each frame · doesn't need to move every frame
- ☐ Fire: 3–4 colours, no outline, stable base, embers = legal orphans · Water: oval chains, crests steeper · Bolt: redraw each frame, 2–3 frames · Wind: swirls, 1 px/frame leaves

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Sheet 18 · The Pixel Studio — tools & pipeline (Lesson 18)

Words on this sheet: **nearest neighbour** = scale with hard square pixels, no blur (a.k.a. hard edges / point / pixelated) · **integer scaling** = whole numbers only (2×, 4×, 8×) · **indexed colour** = pixels store palette numbers · **frame tag** = a named frame range (`walk`) · **pixel-perfect mode** = a pencil that removes corner doubles.

> **P·I·X·L — "Palette small · Integer scale · eXport PNG · Look at 100 %."** This sheet is the I and the X.

Clip Studio Paint EX (Ver. 5 names; older names in brackets):

- ☐ New canvas in px · for animation set timeline **Image interpolation: Hard edges (nearest neighbor)**
- ☐ View > Grid · Grid/Ruler bar settings: Gap 1 px, divisions 1 · Window > Canvas > New Window at 100 %
- ☐ Pen > Marker group > **Dot pen** (1 px, no AA) · other pens: Advanced Tool Settings [Sub Tool Detail] > Anti-aliasing: **None**
- ☐ Fill: Anti-aliasing off + **Area scaling off** · Auto select: AA off, Color margin 0 · Text: AA off
- ☐ Transform: Interpolation method **Hard edges (nearest neighbor)** — always
- ☐ Colour: Window > Color Set > Edit color sets → Add Color (drop a Lospec 1× PNG on a layer and eyedrop)
- ☐ Scale BEFORE export: Edit > Change Image Resolution, 800 %, Hard edges → Export PNG (transparency on)
- ☐ Animation export: Image sequence (PNG) at 1× for engines · Animated GIF for sharing with **Dithering OFF** · no sprite-sheet export (engines take sequences)

ibisPaint X:

- ☐ New canvas → Any Size 32×32 (or Animation)
- ☐ View menu: **Grid** on (Grid 1 Interval 1) · **Display when zooming → Pixelated**
- ☐ Digital Pen 1.0 px · brush parameters > Shape > **Antialias OFF** (eraser too)
- ☐ Bucket: Strength 0 % · Expansion 0 px
- ☐ Transform: Interpolation **Nearest Neighbor (Dot)** · Canvas > Resize: **Nearest Neighbor (Pixel)**
- ☐ Save Image → PNG, Transparent ON · never "Super-resolution x2" · enlarge by Resize, export, keep the 1× original
- ☐ Animation: Save Animation Video to share; save each frame as transparent PNG for a game

Purpose-built editors (pixel-perfect pencil, indexed palettes, ms timing, tags, sprite sheets):

- ☐ Free: **Pixelorama** (MIT; desktop + browser) · **LibreSprite** (GPL fork of 2016 Aseprite) · **Piskel** (browser) · **GraphicsGale** (Windows) · **Pixilart** (browser)
- ☐ Paid standard: **Aseprite** $19.99 — or compile the public source yourself for personal use (legal; no redistributing)

In the engine (pixels stay square):

- ☐ GODOT: Stretch Mode viewport · Aspect keep · Scale Mode **integer** · Default Texture Filter **Nearest** · Snap 2D Transforms to Pixel
- ☐ UNITY: sprites Filter **Point**, Compression **None**, same PPU · **Pixel Perfect Camera** (Upscale Render Texture / Pixel Snapping)
- ☐ UNREAL: texture → Sprite Actions > **Apply Paper 2D Import Settings** · orthographic camera
- ☐ WEB: `image-rendering: pixelated;` · `ctx.imageSmoothingEnabled = false;`
- ☐ One base resolution, integer-scaled as a whole · snap the camera too

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Every sheet is the same sentence in a different costume: cause first → body next → loose parts last → settle. (C·B·L·S — "Cool Bodies Lag Softly.")

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